

Multiple Choice

1. **Answer:** C
Options: A) 42.1 MeV; B) 34.8 MeV; C) 39.2 MeV; D) 40.5 MeV; E) 38.7 MeV
Note: Correct option: C - 39.2 MeV
2. **Answer:** D
Options: A) Mars Science Laboratory; B) Mars Odyssey; C) Mars Atmosphere and Volatile Evolution (MAVEN); D) Phoenix Mars Lander; E) Mars Reconnaissance Orbiter
Note: Correct option: D - Phoenix Mars Lander
3. **Answer:** A
Options: A) a) 180 units, b) 270 units; B) a) -360 units, b) -240 units; C) a) 270 units, b) 180 units; D) a) -180 units, b) 180 units; E) a) -270 units, b) -180 units
Note: Correct option: A - a) 180 units, b) 270 units
4. **Answer:** E
Options: A) rate of change of the magnetic flux through S; B) electric flux through S; C) rate of change of the electric current through S; D) time integral of the magnetic flux through S; E) rate of change of the electric flux through S
Note: Correct option: E - rate of change of the electric flux through S
5. **Answer:** B
Options: A) 2.5 Å; B) 3.86 Å; C) 4.5 Å; D) 1.5 Å; E) 1.24 Å
Note: Correct option: B - 3.86 Å

True / False

6. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'heat energy'.
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'refraction'.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 6.86×10^2 N. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 7.96 m/s^2 and 10.1 s. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key